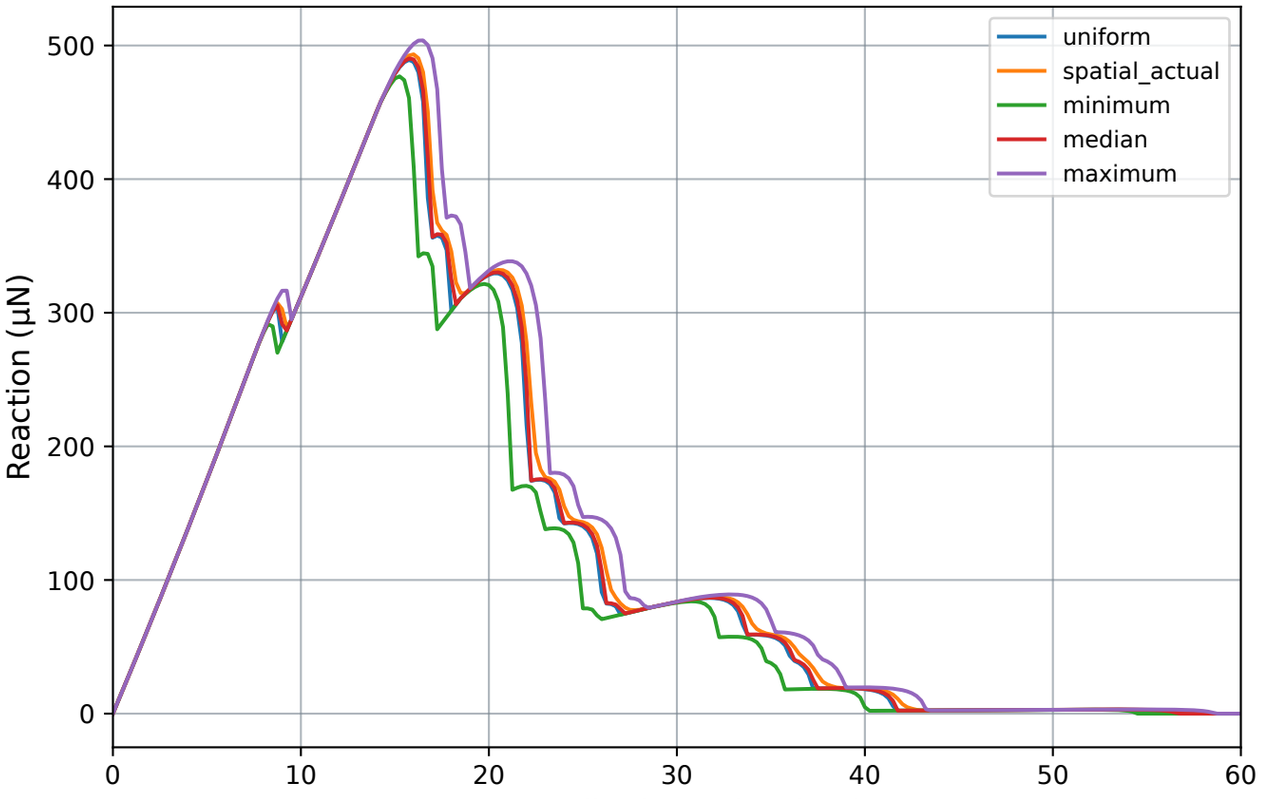
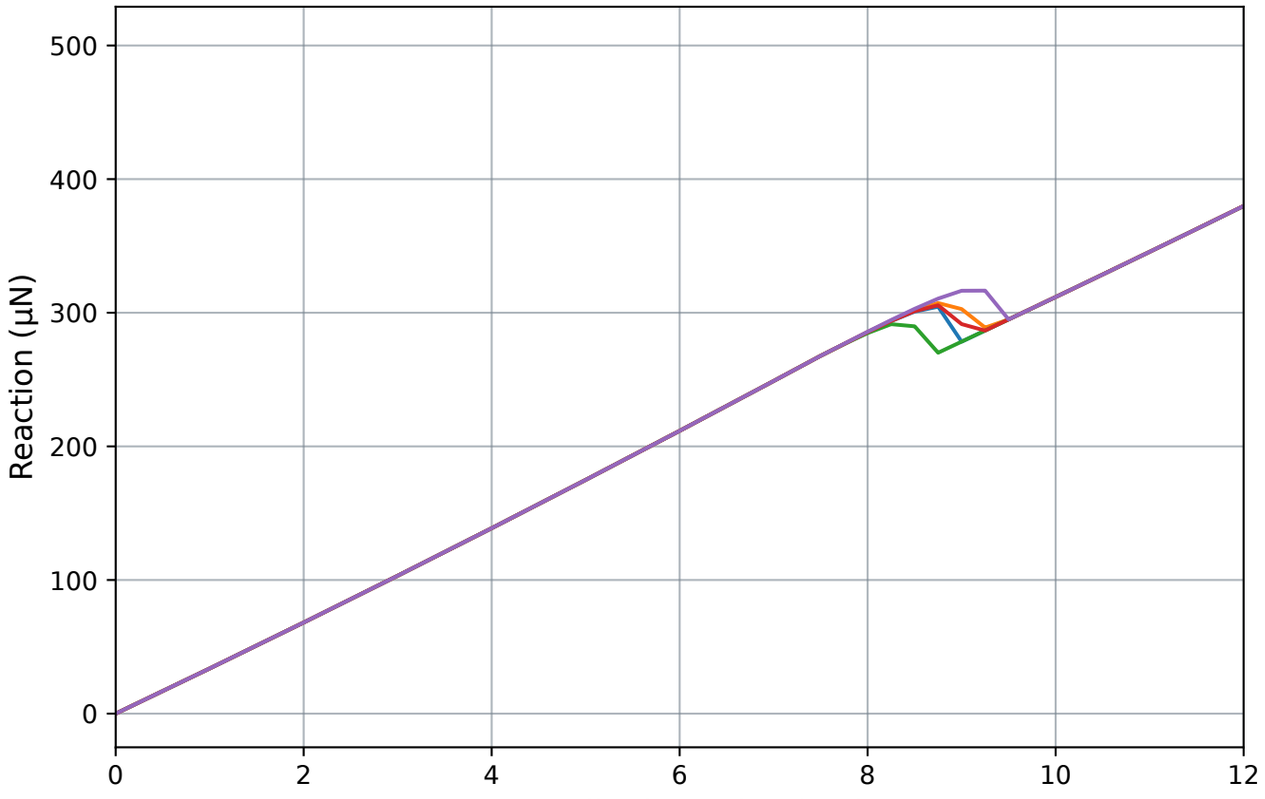
Material-path verification · same complete endpoint-star energy · immutable pair properties

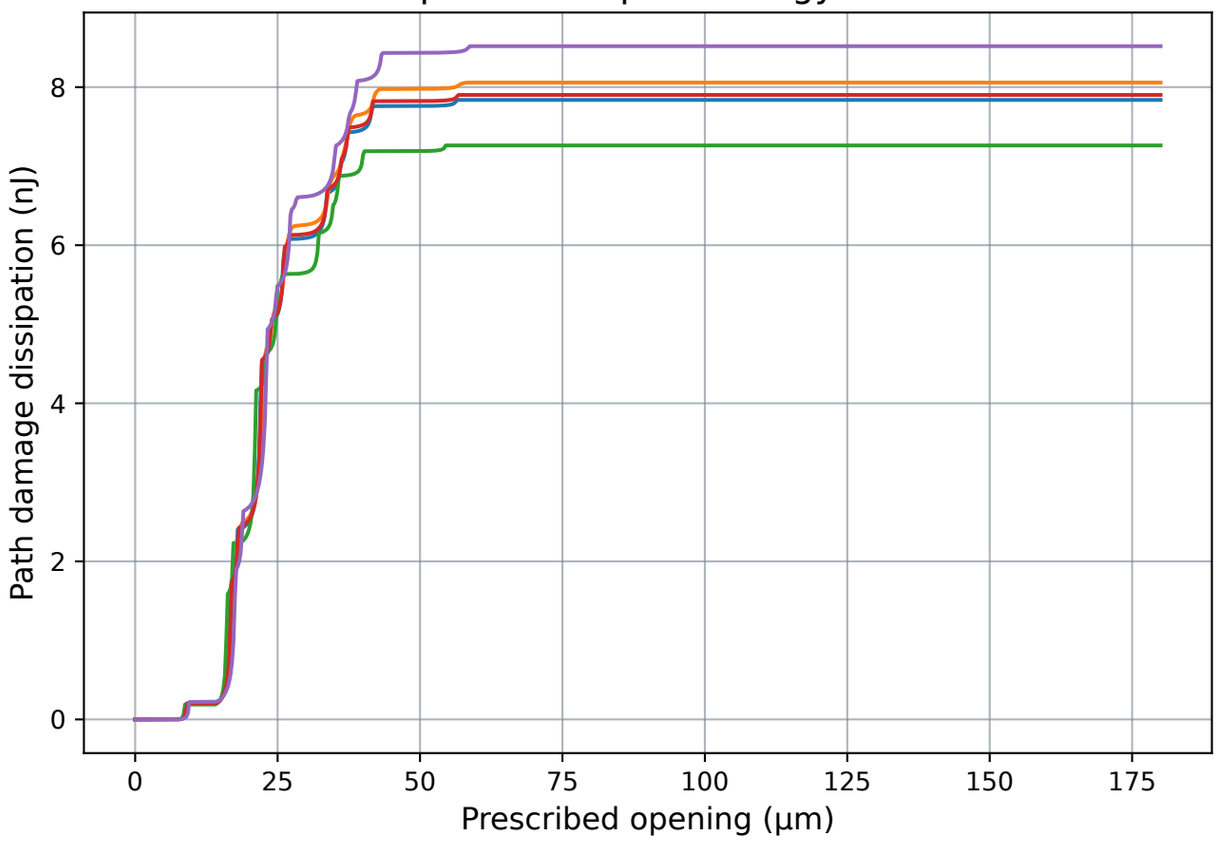
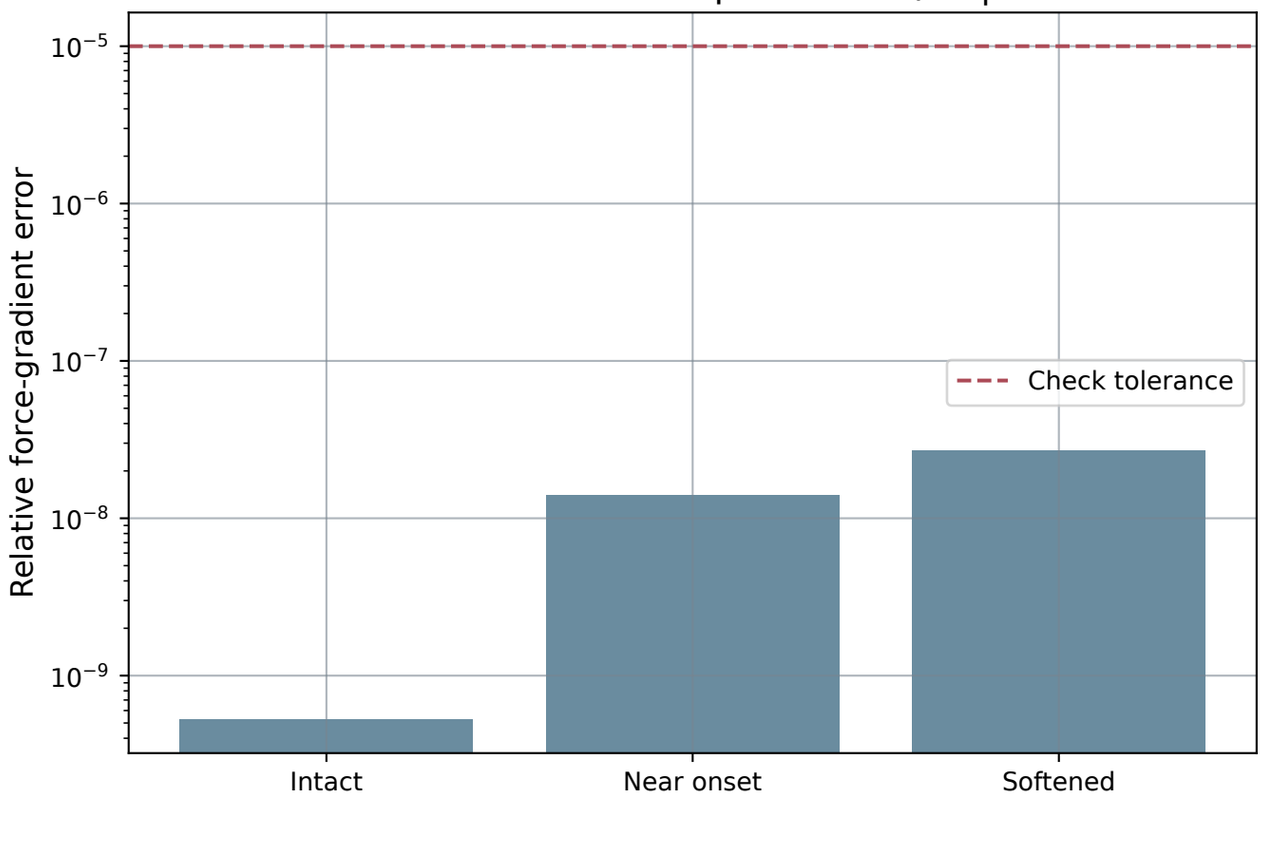
Full discrete opening response



Common elastic response and onset



Offline prescribed-path energy · NOT Gc



Native force vs independent ∂U/∂x | d